MASTER FUNCTIONAL SKILLS

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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

Sample Paper 1 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 60 / 60 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Functional-Skills-Level-2-Sample-Paper-1

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Swimming Pool



The town leisure centre has a swimming pool. The pool is 25 metres (m) long and can be divided into 8 lanes for fitness training and races.

You have been asked to help with some tasks about the swimming pool and the users.

Complete activities 1, 2 and 3 based on the documents provided for each activity.

SAMPLE

Activity 1

Task A

1. The swimming pool is being used for lane swimming. There are 8 lanes in the pool. 3 of the lanes are reserved for **fast swimmers**.

What proportion of the lanes are available for **slower swimmers**?
Show your answer as a **percentage**.

Q1a1 Worked Solution:

Marks available: 3

You must show your working:

62.5 (%) with valid check
5/8 x 100 seen or other valid approach or evidence of (award this mark if 37.5% seen)
1
62.5 (%) accept 63% or 62% if 62.5 seen
1
valid check (not repeat), for example, 62.5 /100 x 8 = 5 or 0.625 x 8 = 5 or use of 3/8 or 0.375 as a check
1
[3 marks]

Your answer:

Show how you can check your answer:

SAMPLE

2. A plan of the swimming pool is required for a poster.

The plan should fit in a 10 centimetres (cm) x 10 cm space and be the maximum size.

What scale should be used to draw the 25 metres (m) x 16.8 m pool?

Q1a2 Worked Solution:

Marks available: 2

You must show your working:

1:250 or 1 cm = 250 cm or 1 cm = 2.5 m (accept 4 mm = 1 m, 5 m = 2 cm, 0.4 cm = 1 m)
10 cm is equivalent to 25 m or 2500 cm or evidence of
1
Correct answer only (CAO) 1 cm = 250 cm or 1 cm = 2.5 m or 1:250 (accept 4 mm = 1 m, 5 m = 2 cm, 0.4 cm = 1 m)
1
[2 marks]

Your answer:

SAMPLE

Task B



1. There is a children's pool which measures 16 m x 8.75 m

The main swimming pool measures 25 m x 16.8 m

What is the ratio of the area of the children's pool compared with the main pool? Show your answer **in its simplest form**.

Q1b1 Worked Solution:

Marks available: 3

You must show your working:

1:3
25 x 16.8 = 420 (m2) and 16 x 8.75 = 140 (m2) or 420 and 140 seen
1
140:420 or 140 to 420 or 140/420 (accept 420:140 or 420/140 seen)
1
1:3 or 1 to 3 (don't accept 3:1 unless it's explicitly clear from final labelling/presentation)
1
[3 marks]

Your answer:

SAMPLE

2. A swimmer has been using a pool with a 50 yard length.

How many lengths would the swimmer need to do in the 25 m length pool to equal 10 lengths in the 50 yard pool?

(1 yard = 0.9144 metres)

Q1b2 Worked Solution:

Marks available: 4

You must show your working:

19 with a valid check (accept 18.288, as minimum (18 isn't acceptable))
2
Activity 1
50 x 0.9144 x 10 or 50 x 0.9144 or 25/0.9144
1
457.2/25 (Follow through (FT) on their 457.2 only) or 500/27.34 (FT on 27.34 only)
1
CAO 19 or 18.288 as minimum e.g. 18.3 but not 18.2 or 18
1
valid check e.g. 19 x 25 = 475, 18.288 x 25 = 457.2
[4 marks]

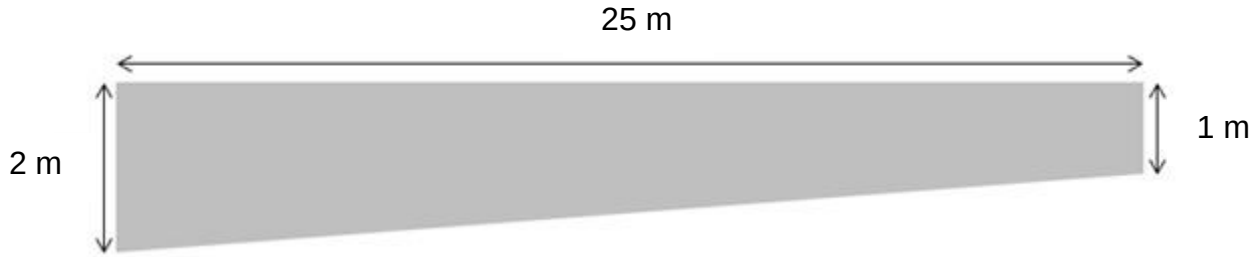
Your answer:

Show how you can check your answer:

SAMPLE

Task C

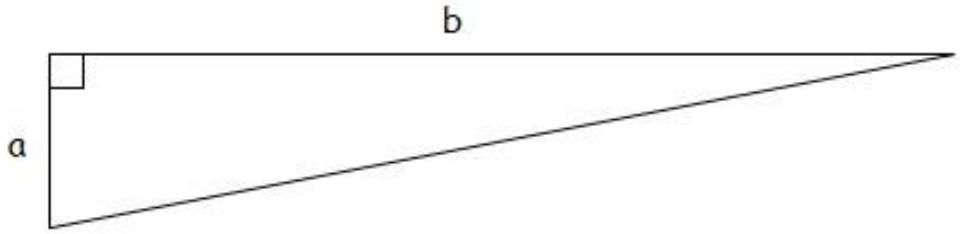
This diagram shows a cross-section of the swimming pool, shown from the side.



(Not drawn to scale)

The swimming pool is 16.8 metres (m) wide.

You may find this information useful:



Area of a right angled triangle = $\frac{a \times b}{2}$

The pool is full to the top edge.

What is the volume of water in the pool? Give your answer **in cubic metres**.

Marks available: 3

You must show your working:

Your answer:

Total marks available: 15

SAMPLE

Activity 2

Task A

1. The tables show the prices for swimming, and the numbers of swimmers expected in the next month.

Type of user	Swimming prices
Adults	£3.80
Children (under 16)	£2.40
Concessions (students, pensioners)	£2.75

Type of user	Expected users for next month
Adults	1250
Children (under 16)	750
Concessions (students, pensioners)	400
Totals	2400

Q2a1 Worked Solution:

£150 profit (units and profit must be shown)

Income = (3.8 x 1250) + (2.4 x 750) + (2.75 x 400) or 4750 + 1800 + 1100 seen

The total running cost for the swimming pool for the next month is expected to be £7500

Based on these figures, what will be the **profit or loss** for the swimming pool in the next month?

Marks available: 3

You must show your working:

1
7650 less 7500 (FT 7650)
1
Made/surplus/profit £150, with units. Must be made clear that this is a surplus, or equivalent explanation/phrasing.
FT but note that 3 marks can only be achieved if '£150 profit' is achieved.
1
[3 marks]

Your answer:

SAMPLE

2. Produce a bar chart to show the **income** from each type of user over the month.

Marks available: 4

Your answer:

Q2a2 Worked Solution:

Correct graph with 3 columns showing 4750, 1800, 1100 or their values

3

Activity 2

Accept 3 columns showing 4750, 1800, 1100 or their income values from 2A1 (must be income)

1

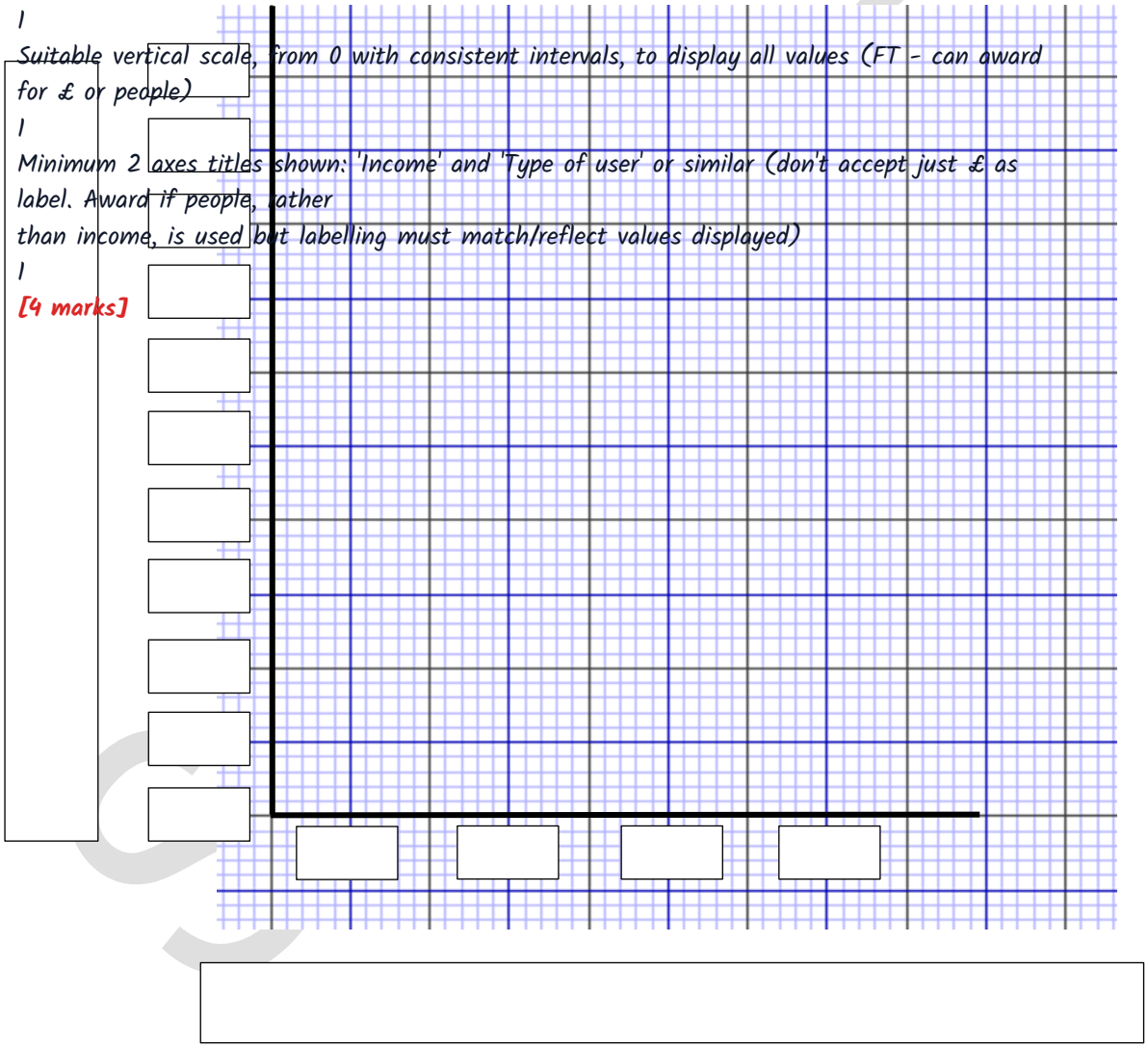
Suitable vertical scale, from 0 with consistent intervals, to display all values (FT - can award for £ or people)

1

Minimum 2 axes titles shown: 'Income' and 'Type of user' or similar (don't accept just £ as label. Award if people, rather than income, is used but labelling must match/reflect values displayed)

1

[4 marks]



Task B

1. The table shows swimming pool water temperatures recorded at the same time each day over a 7 day period. The target temperature is 29 °C

Day	1	2	3	4	5	6	7
Actual Temperature °C	27.9	28.4	29.4	29.2	29.0	28.3	28.0

Q2b1 Worked Solution:

28.6 with valid comment

Find the average (mean) actual temperature over the 7 days.
Show your answer to **1 decimal place**.

What does this tell you about the actual water temperature compared with the target temperature?

Marks available: 3

You must show your working:

27.9 + 28.4 + 29.4 + 29.2 + 29.0 + 28.3 + 28.0 /7 or evidence of
1
CAO 28.6
1
On average, the actual temperature is not reaching temperature/lower/colder than the target temperature (or
equivalent). Do not award without reference to the average/mean/typical temperature being lower. FT on their value.
1
[3 marks]

Your answer:

SAMPLE

2. What is the probability of any 1 of the recorded temperatures being within 0.5 °C above or below the target temperature of 29 °C?

Give your answer as a decimal, shown to **2 decimal places**.

Marks available: 3

Q2b2 Worked Solution:

You must show your working:

0.43
3 temperatures within 0.5 ° C identified (29.4, 29.2, 29.0)
1
CAO 3/7 or evidence of (e.g. 42.86 (%), 3 out of 7, 3 in 7)
1
0.43 Allow FT if their value (/7) is accurately shown to 2 decimal places e.g. 4/7=0.57, 2/7=0.29
1
Total marks:
13
4

Your answer:

[3 marks]

Total marks available: 13

SAMPLE

Activity 3

Task A



1. In a public swimming pool, an event attracts 288 people.

31.25% of these are swimmers and the rest are spectators.

Q3a1 Worked Solution:

How many **swimmers** are there?

Marks available: 2

You must show your working:

90
0.3125 x 288 or equivalent
1
CAO 90
1

Your answer:

[2 marks]

2. There are 128 people who are visiting the swimming pool for the first time.

What proportion of all the 288 people is the number of first time visitors?
Give your answer as a fraction **in its simplest form**.

Marks available: 2

Q3a2 Worked Solution:

You must show your working:

4/9
128/288 or 44.4% or 0.44 or similar showing correct proportion
1
CAO 4/9
1

[2 marks]

Your answer:

SAMPLE

Task B

The table shows the times in seconds for swimmers in 2 clubs in the 100 m freestyle race.

For each club, compare the time for the **fastest swimmer** with the **median time**.

Describe what your results tell you.

Club A Times (seconds)		Club B Times (seconds)
68.2		69.4
68.8		69.5
69.5		69.9
71.5		70.1
72.0		70.5
72.2		71.5
72.9		72.1
73.4		72.8
74.2		73.7
75.1		74.8
75.9		

Marks available: 5

You must show your working:

Your answer:

SAMPLE

Task C

The MET value (Metabolic Equivalent of Task) is a measure of the level of activity.

This table shows some MET values for different activities:

Activity	MET
Cycling	5.0
Gymnastics	5.5
Swimming	6.0
Tennis	7.0
Volleyball	4.0
Walking	3.5

A swimmer weighs 75 kilograms (kg) and swims for **30 minutes**.

Use the formula below to calculate the calories used. Give your answer to the **nearest whole number**.

Calories per minute = MET x weight (kg) x 0.0175

Marks available: 3

You must show your working:

Your answer:

Total marks available: 12

End of assessment

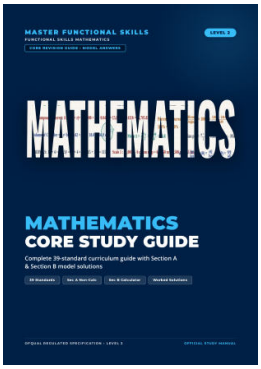
SAMPLE

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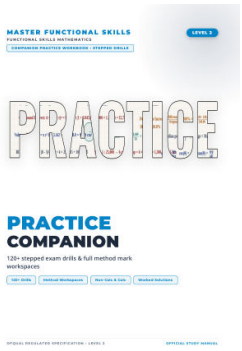


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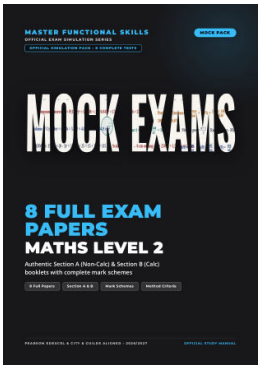


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